Fynn Gerding



ELLIS PhD Researcher in Information-Theoretic Machine Learning

TU Graz, Signal Processing and Speech Communication Laboratory

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Research Profile

PhD researcher at Graz University of Technology working on information-theoretic analyses of neural networks, with a focus on compression, redundancy, generalization, and resource-efficient AI. My work combines probability, statistics, deep learning, and information theory to understand why neural networks generalize and how these insights can support more efficient model design and training.

Research interests: mutual-information estimation, partial information decomposition, redundancy and synergy in overparameterized models, Bayesian and stochastic neural networks, interpretable deep learning, and Green AI.

Research Experience

Graz University of Technology, *ELLIS PhD Researcher* Jun 2026 to present

Signal Processing and Speech Communication Laboratory. Working on the FWF project *Information Planes and Decompositions*, studying compression and redundancy in neural networks with modern information-theoretic tools, including alternatives to classical mutual information and approximations of partial information decomposition.

- Compression and generalization in neural networks
- Redundancy and synergy in overparameterized models
- Mutual-information estimation and approximate partial information decomposition
- Efficient AI practices, including pruning, early stopping, and regularization
- Experiments across fully connected, convolutional, graph, Bayesian, binarized, stochastic, and transformer networks

Radboud University, *M.Sc. Thesis Research* Aug 2024 to Feb 2026

How environment structure shapes reinforcement learning. Extended my B.Sc. thesis from the Penrose P3 tiling to Penrose P2 and the aperiodic monotile, built Gymnasium environments for all three, and developed an algorithm that exploits novelty in Markov chains to improve navigation. Separate project on (Bayesian) neural-network initializations and priors and their effect on convergence stability and robustness.

Education

PhD in Information-Theoretic Machine Learning Jun 2026 to present

Graz University of Technology; *Graz, Austria*

ELLIS PhD Program. Advisors: Bernhard Geiger (TU Graz and Know Center Research GmbH), exchange advisor Giuseppe Durisi (Chalmers University of Technology).

M.Sc. Intelligent Technology (cum laude) Aug 2024 to Feb 2026

Radboud University; *Nijmegen, Netherlands*

B.Sc. Artificial Intelligence (Honors Program, cum laude) Aug 2021 to Jul 2024
Radboud University; *Nijmegen, Netherlands*

Theses

M.Sc. thesis, Radboud University, 2026. Reinforcement learning on aperiodic tilings: Gymnasium environments for Penrose P2, P3, and the aperiodic monotile, and a novelty-based navigation algorithm. *Not publicly available.*

B.Sc. thesis, Radboud University, 2024. *Navigating Aperiodicity: Challenges of Reinforcement Learning.* Advisors: Tal Kachman, Serge Thill. Radboud thesis repository.

Teaching

Radboud University, *Teaching Assistant* Feb 2023 to Jan 2026

Courses: Reinforcement Learning, Functional Programming, Frequentist Statistics, Research Skills, and Bachelor Thesis. Led tutorials, designed and graded assignments and exams, supported student research projects, and coordinated the Bachelor-thesis supervision process.

Industry Experience

Nedap N.V., *Data Scientist* Aug 2023 to Jan 2024

Developed an AI-driven entity-detection system for unstructured clinical text and prototyped a customer-facing chatbot for navigating Nedap's healthcare software suite. Collaborated with product and engineering teams, documented experiments, and supported model evaluation.

SUE B.V., *AIOps Engineer* Jul to Aug 2023

Built a scalable data pipeline to automate the use of large language models in industry settings. Worked with high-volume unstructured text, open-source LLMs, and cloud-based deployment environments, with a focus on robust integration.

Honors and Service

ELLIS PhD Program, admitted student 2026
M.Sc. Intelligent Technology, cum laude, Radboud University 2026
B.Sc. Artificial Intelligence, Honors Program, cum laude, Radboud University 2024
Honors student committee (NSHV), organising formal activities for honors students
Fundraising for Right to Play

Skills

Machine learning: deep learning, Bayesian machine learning, generative models, reinforcement learning, neural-network training and evaluation
Theory: probability, statistics, information theory, mutual-information estimation, generalization analysis
Programming and tools: Python, PyTorch, TensorFlow, Java, Git, experimental ML pipelines
Languages: German (native), English (full professional proficiency), Dutch (elementary)